Deep Learning

Lecture 4: Adversarial models

Chris G. Willcocks

Durham University

Lecture overview

1 Generative adversarial networks

- definition
- properties
- mode collapse
- Lipschitz continuity
- spectral normalisation
- conditional GANs
- information maximizing GANs
- modern GAN training
- adversarial fine-tuning
- adversarial autoencoders

2 Popular applications

- unpaired translation
- super resolution
- adversarial anomaly detection

3 Attacks and detection

- attacks
- deepfake detection

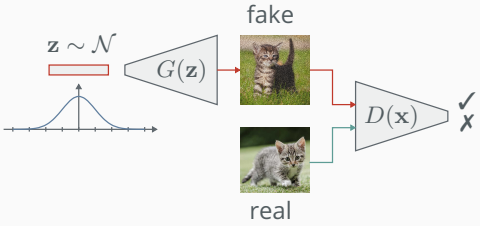
Generative adversarial networks definition

Definition: generative adversarial networks

A generative adversarial network (GAN) is a non-cooperative zero-sum game where two networks compete against each other [1].

One network $G(\mathbf{z})$ generates new samples, whereas D estimates the probability the sample was from the training data rather than G:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))].$$

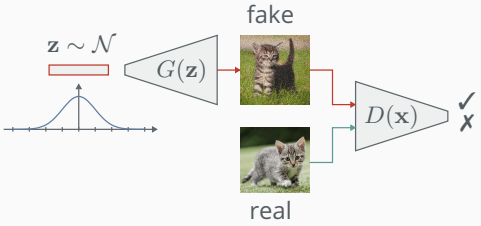


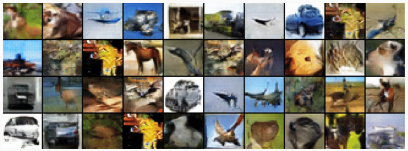
GAN **properties**

GANs benefit from differentiable data augmentation [2] for both reals and fakes, but are otherwise notoriously difficult to train:

- Non-convergence
- Diminishing gradient
- Difficult to balance
- Mode collapse (next slide)

Link to Colab example





Definition: mode collapse

This is where the generator gets stuck producing, or rotates through, a small subset of outputs, and the discriminator is unable to get it out of the trap. Mode collapse is arguably the main limitation of GANs.

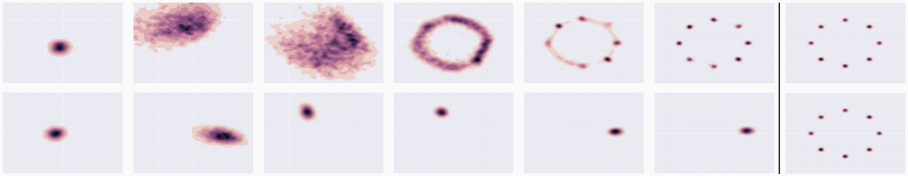


Figure from [3]. The final column shows the target data distribution and the bottom row shows a GAN rotating through the modes.

Definition: Lipschitz function

A function f is Lipschitz continuous if it is bounded by how fast it can change. Specifically if there exists a positive real constant k where:

$$|f(x) - f(y)| \leq k|x - y|,$$

for all x, y in the domain. For example, any function with a bounded first derivative on an interval is Lipschitz on that interval.

Wasserstein GANs [4, 5] use a 1-Lipschitz discriminator (called a critic) while optimising the Wasserstein objective.

Distance functions are 1-Lipschitz

Definition: spectral normalisation

The matrix (spectral) norm defines how much a matrix can stretch a vector $\mathbf{x}$:

$$\|\boldsymbol{A}\| = \max_{\mathbf{x}\neq 0} \frac{\|\boldsymbol{A}\mathbf{x}\|}{\|\mathbf{x}\|}$$

Spectral normalisation [6] divides each weight matrix by its spectral norm $\sigma(\boldsymbol{W})$. For a sequence of linear layers with 1-Lipschitz activations, this bounds the network's Lipschitz constant:

$$\hat{\boldsymbol{W}}_{\text{SN}}(\boldsymbol{W}) = \boldsymbol{W}/\sigma(\boldsymbol{W})$$
$$\sigma(\hat{\boldsymbol{W}}_{\text{SN}}(\boldsymbol{W})) = 1$$
$$\|f\|_{\text{Lip}} \leq 1$$

Spectrally normalised discriminator

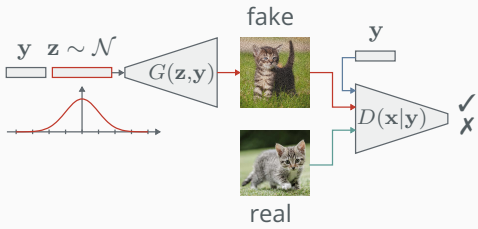
```
class Discriminator(nn.Module):
  def __init__(self, f=64):
    super().__init__()
    self.discriminate = nn.Sequential(
      spectral_norm(Conv2d(1, f, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f, f*2, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*2, f*4, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*4, f*8, 3, 1, 1)),
      nn.LeakyReLU(0.1, inplace=True),
      nn.MaxPool2d(kernel_size=(2,2)),
      spectral_norm(Conv2d(f*8, 1, 3, 1, 1)),
      nn.Sigmoid()
    )
```

Definition: conditional GAN

GANs can be conditioned with labels $\mathbf{y}$ if available [7] by feeding the label information into both the generator and the discriminator:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x},\mathbf{y}\sim p_{\text{data}}(\mathbf{x})}[\log D(\mathbf{x}|\mathbf{y})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z},\mathbf{y})|\mathbf{y}))].$$

Link to Colab example



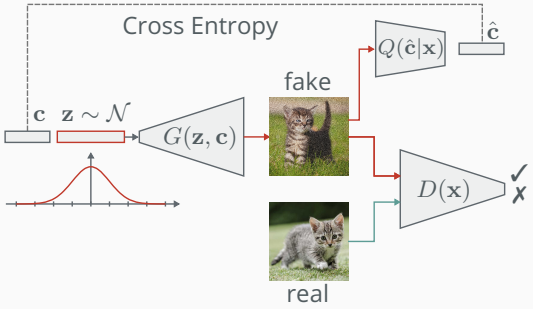
Definition: information maximizing GANs

GANs can be trained to learn disentangled latent representations in a completely unsupervised manner. InfoGAN [8] popularised this by maximizing mutual information between the observation and a subset of the latents:

$$\min_{G,Q} \max_{D} V_{\text{InfoGAN}}(D, G, Q) = V(D, G) - \lambda L_I(G, Q)$$

where $L_I(G, Q)$ is a variational lower bound of the mutual information.

Link to Colab example



Definition: relativistic discriminator

A relativistic discriminator compares real and generated scores C, using a sigmoid σ:

$$D_{\text{pair}}(\mathbf{x}, G(\mathbf{z})) = \sigma(C(\mathbf{x}) - C(G(\mathbf{z}))) \,.$$

R3GAN [9] adds **R1 + R2** penalties on input gradients, encouraging locally flat scores near both real and generated samples.

GAT [10] uses this objective with Transformer networks operating in a pretrained image latent space.

GAT-XL/2: curated ImageNet-256 samples.
Hyun et al., ICML 2026 [10].

Generative adversarial networks adversarial fine-tuning

“Stick a GAN on the end”

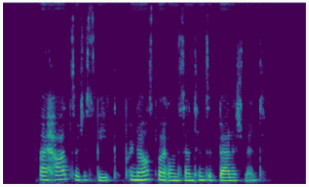
A pretrained generator is further trained with a discriminator to improve the realism of its outputs.

Flow2GAN [11] uses this approach to refine speech from a pretrained flow-matching model, with additional feature-matching and reconstruction losses.

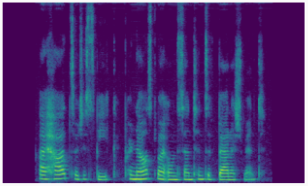
Input: a spectrogram.
Output: a speech waveform.

Listen to the examples

Before adversarial fine-tuning



After adversarial fine-tuning



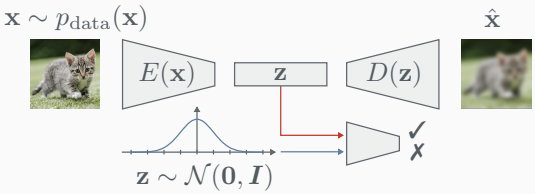
Two sampling steps in both cases.

Yao et al., ICLR 2026, Fig. 2(e–f) [11].

Definition: adversarial autoencoders

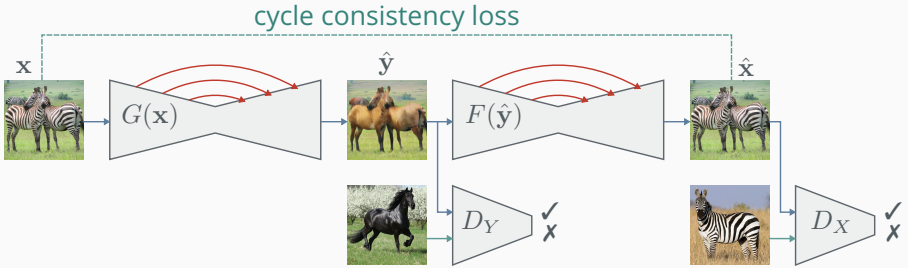
Adversarial autoencoders [12] are generative models that permit sampling.

In addition to the reconstruction loss, such $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$, they use adversarial training to match the aggregated posterior of the hidden code vector $\mathbf{z}$ of the autoencoder with an arbitrary prior distribution, such as $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$.



Definition: unpaired translation (CycleGAN)

CycleGAN [13] propose an adversarial architecture that enables unpaired image translation. It has twin residual generators and two discriminators, which translate between the domains, alongside a cycle consistency loss (an L1 norm) which ensures the mapping can recover the original image.



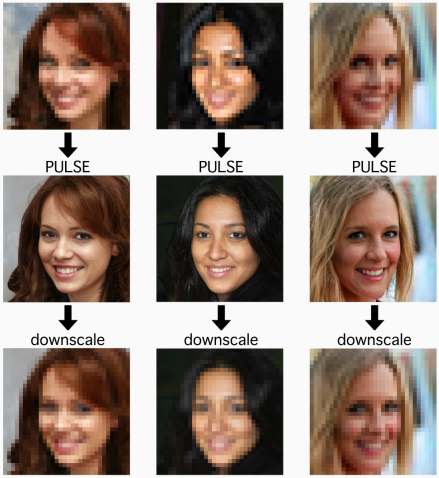
Popular applications super-resolution

Definition: super-resolution

Adversarial models are popular in super-resolution approaches. The challenge is that a single low-resolution (LR) input can map to a distribution of high-resolution (HR) outputs.

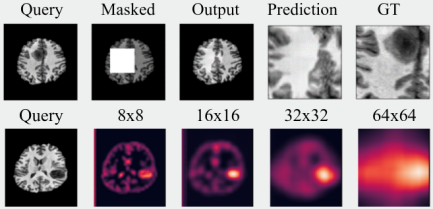
PULSE [14] searches the latent space of a pretrained StyleGAN for outputs matching the LR input after downsampling. It projects latent points onto a hypersphere to keep their radius typical of the Gaussian prior.

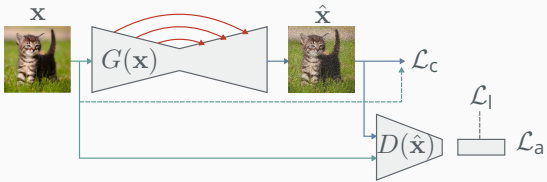
Online example



Definition: anomaly detection

Unsupervised anomaly detectors [15] learn a distribution over normal (healthy) observations. Then, when they observe something not observed in training (unhealthy/dangerous), they may reconstruct it poorly - detecting it as an anomaly. Region-based anomaly detectors [16] learn a distribution over inpainted (erased) regions.



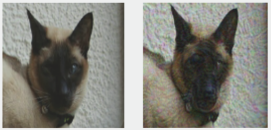


Adversarial examples attacks

Definition: adversarial examples

These are small but intentionally worst-case perturbations that fool the model to output incorrect answers with high confidence [17]. It is possible to generate examples that also fool the human visual system [18]. Cat or dog?

original adv



Example: adversarial examples

Example adding an imperceptibly small vector by the sign of the elements of the gradient of the cost function with respect to the input [17]:

Question: who wins at the limit?

Deepfake generators improve, and detectors learn to spot their remaining artefacts.

If generated and real images have the same distribution [1],

$$p_G = p_{\text{data}} \quad \Longrightarrow \quad D^*(\mathbf{x}) = \tfrac{1}{2}.$$

An image-only detector is then at chance on balanced real/fake samples.

Watermarking

Embed an intentional signal during generation, e.g. **SynthID**, so realistic images can still be detected.

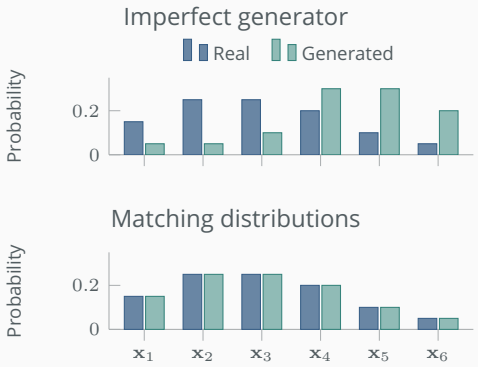


Illustration with six possible images.

References I

[1] Ian Goodfellow et al. **“Generative adversarial nets”.** In: Advances in neural information processing systems. 2014, pp. 2672–2680.

[2] Shengyu Zhao, Zhijian Liu, Ji Lin, Jun-Yan Zhu, and Song Han. **“Differentiable augmentation for data-efficient gan training”.** In: arXiv preprint arXiv:2006.10738 (2020).

[3] Luke Metz, Ben Poole, David Pfau, and Jascha Sohl-Dickstein. **“Unrolled generative adversarial networks”.** In: arXiv preprint arXiv:1611.02163 (2016).

[4] Martin Arjovsky, Soumith Chintala, and Léon Bottou. **“Wasserstein GAN”.** In: arXiv preprint arXiv:1701.07875 (2017).

[5] Ishaan Gulrajani, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron C Courville. **“Improved training of Wasserstein GANs”.** In: Advances in neural information processing systems. 2017, pp. 5767–5777.

[6] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. **“Spectral normalization for generative adversarial networks”.** In: arXiv preprint arXiv:1802.05957 (2018).

References II

[7] Mehdi Mirza and Simon Osindero. **“Conditional generative adversarial nets”.** In: arXiv preprint arXiv:1411.1784 (2014).

[8] Xi Chen, Yan Duan, Rein Houthooft, John Schulman, Ilya Sutskever, and Pieter Abbeel. **“InfoGAN: Interpretable representation learning by information maximizing generative adversarial nets”.** In: Advances in neural information processing systems. 2016, pp. 2172–2180.

[9] Yiwen Huang, Aaron Gokaslan, Volodymyr Kuleshov, and James Tompkin. **“The GAN is dead; long live the GAN! A Modern GAN Baseline”.** In: Advances in Neural Information Processing Systems. 2024. URL: https://arxiv.org/abs/2501.05441.

[10] Sangeek Hyun, Minkyu Lee, and Jae-Pil Heo. **“Scalable GANs with Transformers”.** In: Proceedings of the 43rd International Conference on Machine Learning. Vol. 306. Proceedings of Machine Learning Research. 2026, pp. 49322–49353. URL: https://proceedings.mlr.press/v306/hyun26a.html.

References III

[11] Zengwei Yao et al. **“Flow2GAN: Hybrid Flow Matching and GAN with Multi-Resolution Network for Few-step High-Fidelity Audio Generation”.** In: International Conference on Learning Representations. 2026. URL: https://arxiv.org/abs/2512.23278.

[12] Alireza Makhzani, Jonathon Shlens, Navdeep Jaitly, Ian Goodfellow, and Brendan Frey. **“Adversarial autoencoders”.** In: arXiv preprint arXiv:1511.05644 (2015).

[13] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. **“Unpaired image-to-image translation using cycle-consistent adversarial networks”.** In: Proceedings of the IEEE international conference on computer vision. 2017, pp. 2223–2232.

[14] Sachit Menon, Alexandru Damian, Shijia Hu, Nikhil Ravi, and Cynthia Rudin. **“PULSE: Self-supervised photo upsampling via latent space exploration of generative models”.** In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2020, pp. 2437–2445.

References IV

[15] Samet Akçay, Amir Atapour-Abarghouei, and Toby P Breckon. **"Skip-ganomaly: Skip connected and adversarially trained encoder-decoder anomaly detection".** In: 2019 International Joint Conference on Neural Networks (IJCNN). IEEE. 2019, pp. 1–8.

[16] Bao Nguyen, Adam Feldman, Sarath Bethapudi, Andrew Jennings, and Chris G Willcocks. **"Unsupervised Region-based Anomaly Detection in Brain MRI with Adversarial Image Inpainting".** In: arXiv preprint arXiv:2010.01942 (2020).

[17] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. **"Explaining and harnessing adversarial examples".** In: arXiv preprint arXiv:1412.6572 (2014).

[18] Gamaleldin Elsayed, Shreya Shankar, Brian Cheung, Nicolas Papernot, Alexey Kurakin, Ian Goodfellow, and Jascha Sohl-Dickstein. **"Adversarial examples that fool both computer vision and time-limited humans".** In: Advances in Neural Information Processing Systems. 2018, pp. 3910–3920.

[19] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. **Deep learning.** Available online, MIT press. 2016.